Stage 1: Multi-Source Security Invariant Extraction and Transfer Pipeline

1. Knowledge Corpora & Extraction

Multi-Source Corpora

- Compiler Source & Assertions (GCC/LLVM)
- ISA & ABI Specifications (x86, ARM, RV)
- Historical Bug Patches (e.g. CVE-2023-4039)

Dual Extraction Paths

Segmented CoT Review
Partitioned pass & function boundary slicing

Historical Bug Probe Pool
Distilled root causes across compiler backends

2. Hybrid RAG & Generalization

Dual-Aspect Query Formulation

Sparse BM25: Critical code tokens & ASTs

Dense Vector: Semantic root cause embeddings

Cross-Mechanism Transfer Pipeline

Distill Root Cause → **Filter** Structural → **Specialize** Frame Map

- Transfers across mechanisms (e.g. Canary → PAC)
- Yields candidate invariants for diverse backends

3. Acceptance Gates & Catalog

Triple Acceptance Screening

1. **Grounding Gate:** Source evidence & assertions
2. **Observability Gate:** Binary or runtime check
3. **Novelty Gate:** Deduplication threshold ≥ 85%

Security Invariant Catalog Φ

$\Phi = \{ \phi 1, \phi 2, ..., \phi n \}$
Formal Security Invariant Contracts

- Formalized preconditions & assertions
- Readied for downstream machine oracles

↻ Confirmed Bug Feedback (Iterative Corpus Enrichment)